Elvern Neylmav Tanny

M: +62 81230222663 | Email: elvernneylmav@gmail.com | LinkedIn: in/elvern-neylmav-t | GitHub: github.com/verneylmavt

EDUCATION

Singapore University of Technology and Design (SUTD)
Bachelor of Engineering in Computer Science and Design **Sep 2021 – Oct 2025**

WORK EXPERIENCES

Mekari (Full-Time)
Associate AI Engineer **Indonesia, Jan 2026 – Present**

- Developing and improving RAG workflows using LlamaIndex and LangChain, combining vector search over internal business documents (HR policies, financial schemas, CRM knowledge bases) with live transactional data to improve groundedness and reduce hallucinations in Mekari Airene's responses.
- Optimizing production LLM usage by implementing prompt versioning, caching, and token-cost monitoring, and by benchmarking multiple models (OpenAI GPT family and open-source variants) to balance quality and cost for SaaS workloads.

Amazon Web Services (Part-Time)
Subject Matter Expert (SME) in AI & ML **Remote, Jan 2026 – Present**

- Designing, reviewing, and validating certification exam content and training materials, ensuring alignment with real-world machine learning workflows, including model development, deployment, and MLOps practices on AWS.
- Collaborating with cross-functional AWS teams to evaluate technical accuracy and relevance of AI/ML topics (e.g., SageMaker, data pipelines, model optimization), influencing exam blueprints and learning content to reflect current industry standards.

Ernst & Young (Full-Time)
Technology Consulting Intern **Singapore, Aug 2023 – Dec 2023**

- Participated in the SAP S/4HANA ERP process harmonization phase by collaborating with stakeholders to gather business requirements and standardize workflows for Projects & Program Management (PPM) and Record to Report (RTR) workstreams.
- Translated process requirements into data specifications by defining mapping logic to support the data harmonization phase.

ACADEMIC PROJECTS

SUTD Chatbot (Retrieval-Augmented Generation)
AI/ML/LLM Engineer **Jan 2025 – Apr 2025**

- Built a web scraping and preprocessing pipeline to extract relevant content from ~1K HTML pages on the SUTD website.
- Segmented and embedded the extracted content with SentenceTransformers, storing results in an optimized in-memory vector store.
- Fine-tuned a LLaMA 3.2–3B model using LoRA PEFT on a 200-pair QA dataset, hosted the model on an Amazon SageMaker endpoint, and integrated the model into a LangGraph RAG pipeline served via a FastAPI application on AWS Fargate.
- Implemented an automated LLM-as-a-judge evaluation, achieving ~4.8/5.0 rating in accuracy, relevance, and groundedness.

Blood Sugar Prediction (Deep Neural Network)
Data Scientist **Jan 2025 – Apr 2025**

- Performed a multi-stage preprocessing pipeline, including data cleaning (linear interpolation, zero/mean imputation), data augmentation (Gaussian noise injection), and feature engineering (rolling statistics), leading to a 15% decrease in RMSE.
- Developed a Temporal Convolutional Network (TCN) based on "*Temporal Convolutional Networks for Action Segmentation and Detection (Lea et al., CVPR 2017)*" paper, utilizing causal convolutions and residual connections to capture temporal dependencies.

OOG Navigator - Cargo Management System (Web App)
Data Scientist & Back-End Developer **Sep 2024 – Apr 2025**

OOG Navigator is a cargo management web app, in collaboration with DB Schenker, aimed at optimizing the workflow for out-of-gauge (OOG) cargo.

- Developed, trained, and tuned an XGBoost ML model for equipment recommender microservice using Amazon SageMaker Training and Hyperparameter Tuning Jobs, with the model inference deployed via AWS Lambda.
- Architected and deployed a Node.js back-end system on Amazon ECS, implementing API endpoints and business logic to coordinate interactions among microservices (document analyzer, template engine, equipment recommender) on AWS Lambda, database on Amazon Aurora, and storage on Amazon S3.

Sentiment Analysis (Deep Neural Network)
Data Scientist **Sep 2024 – Dec 2024**

- Utilized FastText embeddings to handle out-of-vocabulary (OOV) words, improving model generalization over static embeddings.
- Developed a hybrid CNN-RNN architecture with attention mechanisms using PyTorch, integrating convolutional feature extraction, recurrent sequence processing, and context awareness to enhance the model's semantic understanding.
- Optimized the model hyperparameters (e.g., kernel size) with Optuna, resulting in ~95% accuracy on the IMDB dataset.
- Converted the model to ONNX format and applied optimization and quantization techniques to accelerate inference by 50%.

Invoice Management System (Web App)
Full-Stack Developer **May 2023 – Aug 2023**

An invoice management web app aimed at automating the manual handling of invoices, featuring OCR-based extraction and search functionality.

- Created the front-end "Invoice Upload" module in Svelte, featuring drag-and-drop file selection and data editing functionalities.
- Built the back-end "OCR Request" service in SvelteKit, for automated real-time data extraction and processing.

ADDITIONAL INFORMATION

- Technical Skills: Python (FastAPI, NumPy, Pandas, Matplotlib, Scikit-Learn, XGBoost, PyTorch, TensorFlow, Transformers, LangChain), Spark, Hadoop, HTML, CSS, JavaScript (React, Node.js), SQL, NoSQL, Git, Bash, Docker, Kubernetes, AWS
- Certifications: AWS Certified Cloud Practitioner - Foundational, AWS Certified AI Practitioner - Foundational, AWS Certified Machine Learning Engineer - Associate, AWS Certified Machine Learning - Specialty